Literature Review – Smart Pet Feeder

Student ID: 2408599

Name: Adarsha Prasai

Section: L6CG8

Supervisor: Sangay Lama

Reader: Subash Bista

Table of Contents

1. Introduction 1

2. Literature Review 2

2.1. Design of Smart Pet Food Dispenser using Embedded System 2

- Results 2
- Limitations 3

2.2. Implementation of an IoT Based Pet Care System 3

- Results 4
- Limitations 4

2.3. IoT and Cloud-based Automated Pet Care System 5

- Results 5
- Limitations 6

2.4. Modular Pet Feeding Device 6

- Results 6
- Limitations 7

2.5. Automatic Pet Feed System Applying IoT 7

- Results 8
- Limitations 8

2.6. IoT Based Automatic Pet Feeder 9

- Results 9
- Limitations 10

2.7. Design and Development of a Smart Pet Feeder with IoT and Deep Learning 11

- Results 11
- Limitations 12

2.8. Overall Gap Analysis 12

- Common Gaps: 12

3. Proposed System 13

3.1. Concept Overview 13

3.2. System Architecture and Module Interactions 13

3.2.1. ESP32 Control Unit 13

3.2.2. HX711 Load Cell Amplifier 13

3.2.3. Servo Motor Dispensing Module 14

3.2.4. IoT Communication and Cloud Integration 14
3.2.5. Web Application and Multi-Caregiver Control 14
3.2.6. Offline Synchronization and Analytics 14

3.3. Security, Audit, and Human Oversight 15

4. Technology Overview 15

4.1. HX711 Load Cell Amplifier 15
4.2. Servo Motor 15
4.3. ESP32 Microcontroller 15
4.4. MQTT vs HTTP REST Communication 16
4.5. Cloud Integration 16
4.6. Web Application and Logging Mechanisms 16
4.7. IoT Security and Authentication 16
4.8. Analytics and Visualization 16

5. Research Gap and Motivation 17

6. Conclusion 18

7. References 19

1. Introduction

The Internet of Things (IoT) has enabled pervasive interconnectivity across devices, allowing automation to extend into domestic and companion-animal environments. Within the pet care ecosystem, smart feeders offer scheduling convenience, portion consistency, and remote control. However, despite commercial success, technical limitations persist in most systems. Existing feeders typically use time-based control or uncalibrated rotation, with minimal regard for actual dispensed weight. Consequently, feeding errors, overdispensing, and lack of supervision undermine reliability, especially during owner absence.

Automation in pet care contributes not only to convenience but also to animal health, enabling consistent feeding patterns and mitigating human error. Yet the underlying technical challenge remains: translating mechanical actuation into verified nutritional outcomes. In open-loop feeders, servo motors dispense based on pre-timed durations rather than feedback from actual food weight.

Security, auditability, and resilience are therefore essential qualities of next-generation smart feeders. From a system perspective, these features demand a robust embedded control loop, a synchronized cloud architecture, and a human-in-the-loop safety model. The proposed *IoT-Enabled Closed-Loop Smart Pet Feeder with Multi-Caregiver Web Application and Analytics* aims to unify these dimensions. By combining precise load-cell-based control, secure cloud integration, and an offline-first design, it provides verifiable automation even in intermittent network environments.

2. Literature Review

2.1. Design of Smart Pet Food Dispenser using Embedded System

- **Name of the paper:** Design of Smart Pet Food Dispenser using Embedded System
- **Type of paper:** Conference paper (IEEE Proceedings)
- **Pages:** 1–6
- **Authors:** E. Jain, S. Sharma, R. Agarwal, and N. K. Gupta
- **University:** Symbiosis Institute of Technology, Pune, India
- **Relation:** Introduces a microcontroller-based automatic pet feeder using Arduino and basic IoT integration; demonstrates open-loop control without verification forms baseline mechanical automation that my FYP expands into closed-loop accuracy with analytics.
- **Where found:** IEEE Xplore

(Jain, et al., 2023)

- **Results**

The prototype successfully dispensed food at user-defined intervals and was remotely triggerable through the Firebase-connected app. The authors reported correct motor actuation and stable Wi-Fi control within a local LAN range. However, they did not quantify weight accuracy or latency metrics. User tests were demonstrated qualitatively by showing dispensing at scheduled times. No numerical performance evaluation (such as portion error or throughput) was presented. Therefore, results remain proof-of-concept level rather than validated metrics.

- **Limitations**

The system is open-loop: it does not verify the actual mass dispensed against the target. The Firebase link provides no security or audit trail, and the mobile interface lacks multi-user access control. Offline functionality is absent since the ESP8266 cannot queue commands without network. The design relies on real-time clock schedules hard-coded in firmware, making updates inflexible. No human-in-the-loop safety or notification logic exists, and feed errors cannot be detected autonomously.

2.2. Implementation of an IoT Based Pet Care System

- **Name of the paper:** Implementation of an IoT Based Pet Care System
- **Type of paper:** Conference article (IEEE)
- **Authors:** Y. Chen and M. Elshakankiri
- **University:** University of Regina
- **Relation:** Proposes an IoT edge-based pet monitoring system using Raspberry Pi and AWS; focuses on latency improvement, not feed accuracy relates to my review by demonstrating MQTT-based IoT control and showing lack of closed-loop reliability.
- **Where found:** IEEE Xplore

(Chen & Elshakankiri, 2020)

- **Results**

The system demonstrated low-latency control under edge processing (~50–80 ms average response). The authors reported successful remote video streaming and feeding activation. Quantitative metrics focused on network delay and throughput but not feeding accuracy. The solution showed that edge-assisted IoT can reduce delay compared to pure cloud architecture. However, no testing of portion precision or device autonomy was performed.

- **Limitations**

Hardware cost was high due to Raspberry Pi and AWS use. The system depends on constant Internet availability; no offline operation or recovery logic exists. Security measures were limited to AWS credentials and did not cover end-to-end device authentication or auditing. The architecture is general pet care, not specialized in precise feeding. Data persistence and multi-user roles were not addressed. Furthermore, the design did not measure real-world feed outcome against desired target values.

2.3. IoT and Cloud-based Automated Pet Care System

- **Name of the paper:** IoT and Cloud-based Automated Pet Care System
- **Type of paper:** Journal article (IEEE Access / digital proceedings)
- **Authors:** M. Rajani Devi, V. Jyothi, and D. Nagajyoti
- **University:** Jawaharlal Nehru Technological University, Hyderabad, India
- **Relation:** Describes a cloud-connected feeding and watering system controlled by Blynk and ThingSpeak; demonstrates successful scheduling but lacks mass verification, supports my argument for integrating closed-loop sensing and offline capability.
- **Where found:** IEEE Xplore

(Devi, et al., 2022)

- **Results**

The device worked reliably for remote commands through the Blynk interface and maintained connectivity with the ThingSpeak cloud. The authors reported successful data upload of feed events and pet presence detection. However, quantitative results such as error percentage of food dispensed or latency values were not included. The results are thus functional verifications rather than statistical analysis.

- **Limitations**

Key limitations include lack of closed-loop feedback, no weight verification, and dependency on external cloud servers for core functionality. Offline operation fails if Wi-Fi disconnects. Data is visible only to the account holder; no multi-user role support exists. Security and data integrity are not discussed. The architecture is single-point of failure with limited fault recovery.

2.4. Modular Pet Feeding Device

- **Name of the paper:** Modular Pet Feeding Device
- **Type of paper:** Student project research report / institutional publication
- **Authors:** V. C. M., P. Rao, S. Kiran, and A. Fernandes
- **University:** Dayananda Sagar College of Engineering, Bengaluru, India
- **Relation:** Focuses on hardware modularity of feeders without networking or analytics; forms the mechanical foundation that my FYP builds upon by adding IoT feedback, web dashboards, and authentication.
- **Where found:** Google Scholar

(M., et al., 2023)

- **Results**

Testing showed that the device could dispense food consistently when manually triggered. Measurements of portion uniformity were limited to visual inspection. The authors reported “accurate dispensing” qualitatively but did not provide numeric error data or sensor readings. The device ran on a 12 V DC supply and could operate autonomously without a network.

- **Limitations**

The system lacks any smart functionality or communication layer. It cannot be remotely controlled or monitored, nor store historical feed records. Sensor precision is insufficient for accurate mass measurement. Without a backend or app, multi-user interaction and auditing are impossible. The authors did not address security, calibration, or human-in-loop features

2.5. Automatic Pet Feed System Applying IoT

- **Name of the paper:** Automatic Pet Feed System Applying IoT
- **Type of paper:** Journal article (Modelling, Measurement and Control D)
- **Pages:** 87–93
- **Authors:** A. V. C. Rangel, I. U. Ponce, and I. Soto
- **University:** Instituto de Ingenieria y Tecnologia
- **Relation:** Presents an IoT feeder with HX711 sensor and ThingSpeak cloud logging; lacks feedback control or authentication, directly aligns with my review focus on transforming one-way data collection into verifiable closed-loop systems.
- **Where found:** Measurement and Control D (International Journal)

(Rangel, et al., 2024)

- **Results**

The prototype achieved stable network connectivity and successfully recorded feed data on ThingSpeak. The authors confirmed accurate logging of feeding instances and real-time updates. However, they omitted quantitative accuracy measures, no error analysis between intended and actual dispensed mass or timing variance. Testing was limited to short demonstrations under stable Wi-Fi conditions.

- **Limitations**

Major weaknesses include lack of closed-loop verification, total dependence on connectivity, and unaddressed security. The system cannot adapt to motor variation or grain density differences. Blynk and ThingSpeak impose external dependency and poor scalability. There is no human oversight, error recovery, or multi-user model.

2.6. IoT Based Automatic Pet Feeder

- **Name of the paper:** IoT Based Automatic Pet Feeder
- **Type of paper:** Journal article (International Journal of Scientific Research in Science, Engineering and Technology – IJSRSET)
- **Pages:** 570–577
- **Authors:** R. Robert, C.Jerisha, S.Mable Vimala, N.S. Nanthini, and T.Saranya
- **University:** Annai Vailankanni College of Engineering
- **Relations:** Demonstrates timed servo control for feeding through Blynk with no verification or data persistence; serves as an early open-loop reference, highlighting why my FYP's feedback and audit systems are necessary.
- **Where found:** IJSRSET (Volume 9, Issue 8)

(R.Robert, et al., 2021)

- **Results**

Functional testing verified that feeding actions occurred at scheduled times, with reported sub-second response to mobile commands. The system demonstrated consistent motor rotation but lacked numeric validation of portion consistency. The authors documented photos of successful feeding as qualitative evidence.

- **Limitations**

This system cannot adjust or verify dispensed amounts, has no fault tolerance, and depends entirely on Internet access. There are no authentication or audit mechanisms. Calibration, measurement, and user control granularity are minimal. Offline operation, analytics, or multi-caregiver control are absent.

2.7. Design and Development of a Smart Pet Feeder with IoT and Deep Learning

- **Name of the paper:** Design and Development of a Smart Pet Feeder with IoT and Deep Learning

- **Type of paper:** Conference paper (MDPI Proceedings of 8th International Electronic Conference on Sensors and Applications)

- **Authors:** Oscar E. Castillo-Arceo, Ricardo U. Ruiz Flores, and Patricia C. Santana Mancilla

- **University:** Universidad de Colima, Mexico

- **Relation:** Integrates IoT hardware with YOLOv5-based pet recognition; strong AI use but lacks multi-user control or audit trail, supports my argument for extending smart feeders into secure, transparent IoT ecosystems.

- **Where found:** MDPI Proceedings

(Castillo-Arceo, et al., 2024)

- **Results**

The authors achieved functional pet recognition and portion dispensing aligned to pet identity. They claimed accurate pet differentiation and stable HX711 readings. However, they omitted numerical accuracy or timing results. Performance metrics such as detection confidence or feed error were absent.

- **Limitations**

Despite AI integration, the feeder operated as a self-contained device with no authenticated cloud control or multi-user oversight. There was no audit log or notification mechanism. Dependence on environmental lighting could reduce detection reliability. Offline behaviour, data reconciliation, and user verification were untested.

2.8. Overall Gap Analysis

- **Common Gaps:**
  - **Absence of closed-loop control:** Only two works mention load cells, and neither performs continuous feedback correction or quantitative evaluation of mass error.
  - **Lack of security and auditability:** Every reviewed system except Castillo-Arceo et al. operates without authentication, encryption, or verified logs. Firebase or Blynk connections dominate, offering convenience but no durable record integrity.
  - **Network dependency:** Offline resilience is universally absent. Systems fail entirely without Wi-Fi, indicating no local autonomy or synchronization strategy.
  - **Single-user limitation:** None support multi-caregiver models or role-based access. Pet management remains isolated to a single mobile user.
  - **No human-in-the-loop verification:** Automation occurs blindly with no acknowledgment, override, or feedback channel.

3. Proposed System

3.1. Concept Overview

The conceptual framework envisions an IoT system composed of interlinked hardware, firmware, and software modules operating in a closed-loop configuration. At its core, an **ESP32 microcontroller** interfaces with an **HX711 load cell amplifier** and **servo motor** to dispense precise feed portions. Data communication occurs through a hybrid model of **local control** and **cloud synchronization**, allowing the system to function offline and reconcile data when reconnected. The web application serves as the supervisory layer, hosting multi-user access, analytics, and audit logging.

(Conceptual Architecture Diagram Placeholder)

3.2. System Architecture and Module Interactions

3.2.1. ESP32 Control Unit

The ESP32 serves as the central processing and communication hub, managing sensor input, actuator control, and network communication (Whitepaper, 2024). It continuously reads load cell data via the HX711 ADC and applies a feedback algorithm to maintain target feed mass. When network connectivity is unavailable, data is cached locally in flash memory and queued for transmission upon reconnection.

3.2.2. HX711 Load Cell Amplifier

The HX711 measures the real-time weight of dispensed feed with high precision. By embedding this sensor into a feedback loop, the system monitors the actual mass dispensed and terminates feeding once the desired threshold is reached (Team, 2024) Calibration coefficients are stored in non-volatile memory, ensuring repeatable accuracy.

3.2.3. Servo Motor Dispensing Module

The servo controls a gate mechanism that regulates pellet flow. The ESP32 modulates servo angle in response to HX711 readings. This constitutes a proportional feedback mechanism where dispensing continues until the error between target and actual weight approaches zero (Staff, 2025).

3.2.4. IoT Communication and Cloud Integration

The system employs either MQTT or RESTful HTTP for data transmission. MQTT provides asynchronous, low-overhead updates ideal for constrained IoT devices, while REST API offers transactional reliability for audit events (Rahman, 2024). A cloud backend—integrated per best practices for resilient IoT architecture (Team, 2025) stores feeding logs, calibration parameters, and user actions.

3.2.5. Web Application and Multi-Caregiver Control

The web app, developed using a modern frontend framework (e.g., React or Next.js), interacts through secure APIs. Authentication follows lightweight, mutual verification protocols suitable for IoT ecosystems (Shehu, et al., 2025). Role-based access control allows multiple caregivers to manage schedules collaboratively while preserving accountability. Event logging adheres to principles outlined in (Team, 2025) ensuring immutability and traceability of all interactions.

3.2.6. Offline Synchronization and Analytics

During network loss, feeding records and control events are timestamped locally. Upon reconnection, the device compares local and server logs, resolving conflicts using idempotent tokens to prevent duplication. Analytics modules in the web app visualize feeding frequency, average portion mass, and deviation trends, following practices from (Team, 2025).

3.3. Security, Audit, and Human Oversight

Authentication, encryption, and event logging form the trust backbone. Following guidance from (Team, 2023) and (Wu, et al., 2020) the system employs HTTPS with token-based session validation. All feed events are auditable and reversible by authorized users. Notification mechanisms alert caregivers of anomalies or feeding completion, ensuring human-in-loop supervision.

4. Technology Overview

4.1. HX711 Load Cell Amplifier

The HX711 is a 24-bit analog-to-digital converter optimized for weight sensing. It amplifies microvolt-level signals from the load cell, enabling detection of minute mass variations. In this system, it provides real-time feedback for the dispensing servo, forming the foundation of the closed-loop control algorithm (Team, 2024).

4.2. Servo Motor

A micro servo motor translates control signals from the ESP32 into mechanical movement for the feeding gate. Pulse-width modulation (PWM) defines its angular position. When paired with the HX711 feedback, the servo dynamically adjusts to maintain dispensing accuracy within a predefined tolerance band (Staff, 2025).

4.3. ESP32 Microcontroller

The ESP32, with integrated Wi-Fi and dual-core processor, handles sensor readings, feedback calculations, and network communications concurrently. Its embedded flash storage enables offline data caching and synchronization on reconnection (Whitepaper, 2024).

4.4. MQTT vs HTTP REST Communication

MQTT supports lightweight, bidirectional communication suitable for real-time IoT telemetry, whereas REST ensures structured, authenticated requests (Rahman, 2024) . This system proposes MQTT for live feed-state updates and REST for log synchronization and user transactions.

4.5. Cloud Integration

Cloud services maintain a persistent data layer, following principles from (Team, 2025). The architecture leverages serverless APIs for device communication and NoSQL storage for rapid feed log queries.

4.6. Web Application and Logging Mechanisms

A React-based dashboard interfaces with backend APIs for control and visualization. It implements detailed logging-login attempts, schedule modifications, and feed completions, based on the design guidelines in (Team, 2023) and (Team, 2025).

4.7. IoT Security and Authentication

The project employs mutual authentication and token-based validation to protect device-cloud sessions (Title: "Authentication Schemes for Internet of Things (IoT) Networks"). All communication uses HTTPS with certificate pinning and salted hash storage for credentials, adhering to recommendations in (Wu, et al., 2020).

4.8. Analytics and Visualization

Data analytics modules use aggregated feed data to compute average portion accuracy, caregiver activity frequency, and trend analysis. Visualization dashboards follow frameworks described in (Team, 2025).

5. Research Gap and Motivation

Existing smart pet feeders demonstrate limited technical maturity. Most operate under open-loop control, dispensing food for fixed durations without verifying actual output mass. This results in inconsistent feeding and potential overfeeding issues particularly critical for diet-sensitive animals. While prior designs such as those surveyed in (Company, 2025), indicate increasing automation, they largely prioritize convenience over accuracy or transparency.

Connectivity dependence poses further risks. Feeders relying exclusively on continuous Wi-Fi cannot operate during network disruptions, leaving pets unfed. Existing systems rarely implement offline caching or synchronization models (team & Oliynyk, 2024). Consequently, IoT-based pet feeders remain vulnerable to both communication failures and data inconsistency.

From a systems perspective, these gaps represent deficiencies in **feedback control**, **data governance**, and **operational resilience**. The proposed project directly addresses these through:

- **Closed-loop feedback control** for verified portion accuracy using HX711 mass readings.
- **Secure, multi-caregiver web platform** supporting authenticated roles and auditable logs (Team, 2025)
- **Offline-first synchronization** ensuring functional continuity under network loss (Team, 2025)
- **Integrated analytics dashboard** offering measurable performance feedback and feeding trends (Team, 2025).

6. Conclusion

This research review established a conceptual framework for a next-generation IoT-enabled smart pet feeder emphasizing precision, security, and collaborative management. The system architecture integrates a closed-loop feedback mechanism, an authenticated multi-caregiver web platform, and an analytics-driven control layer.

Compared to existing feeders that rely on static timing and single-user mobile control, this conceptual system embodies a holistic approach, linking embedded hardware intelligence with secure cloud orchestration. The integration of the HX711-based feedback control introduces quantitative reliability, while the offline-first synchronization model addresses real-world connectivity constraints.

The project's long-term vision extends toward intelligent behavioral analytics and adaptive feeding through AI integration (M., et al., 2023).

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